

FLUXNET Annual Paper 2026 — Authorship Report

Prepared: 2026-05-12 | **Updated:** 2026-05-12 (presence file regenerated — see §Methodology update) **Rubric locked:** 2026-05-07 (team meeting) | **Snapshot:** Apr-28 2026 (716 sites) | **Reference year:** 2026

The authorship rubric was locked at the team meeting on 2026-05-07. This report documents the locked numbers, one open caveat, and one pending followup task that must be resolved before the author list is finalised. The headline increased from 2,877 (initial May-12 run) to **2,905** following presence file regeneration with a broader all-variable definition (see §Methodology update below).

Locked rubric

Three latency tiers × five data-volume bins, fully monotone, author counts 2–8:

	>3 yr latency	≤3 yr latency	≤2 yr latency
<5 years	2	3	4
6–10 years	3	4	5
11–15 years	4	5	6
16–20 years	5	6	7
≥21 years	6	7	8

Latency = 2026 – max_data_year per site. Data-volume bins count distinct years with any non-NA flux variable (NEE_VUT/CUT, GPP/RECO_NT/DT × VUT/CUT, LE_F_MDS, H_F_MDS) present in the presence file. See §Methodology update.

Methodology update — Presence file regenerated (2026-05-12)

The presence file (data/snapshots/site_year_data_presence.csv) was regenerated on 2026-05-12 with a broader, all-variable definition, replacing the earlier NEE_VUT_REF-only indicator.

Old definition: a site-year was “present” if NEE_VUT_REF was non-NA in at least one month. This excluded years where only energy or alternative-USTAR-method flux data were available.

New definition: a site-year is “present” if any of the following 12 columns is non-NA in at least one month: NEE_VUT_REF, NEE_CUT_REF, GPP_NT_VUT_REF, GPP_DT_VUT_REF, GPP_NT_CUT_REF, GPP_DT_CUT_REF, RECO_NT_VUT_REF, RECO_DT_VUT_REF, RECO_NT_CUT_REF, RECO_DT_CUT_REF, LE_F_MDS, H_F_MDS.

Coverage change: The old file covered 672 sites (with 88 of the 716 snapshot sites on a snapshot-span fallback). The new file covers all 716 sites directly from the presence data — the fallback is no longer needed and has been removed from the authorship script.

44 sites recovered. These sites were invisible to the NEE_VUT_REF-only indicator because NEE was either CUT-only (VUT statistically infeasible) or NEE processing failed downstream, but the sites have valid energy or GPP/RECO data. This is a methodologically important shift: these are not fringe sites. They include some of the longest-running towers in the AmeriFlux network:

Site	Network	years_of_data (new)	Note
US-NR1	AMF	28	Niwot Ridge — continuous since 1998, NEE CUT-only

Site	Network	years_of_data (new)	Note
US-Ho2	AMF	27	Howland Forest West — VUT infeasible, NEE CUT-only
US-WCr	AMF	24	Willow Creek — 24 yrs GPP/RECO/LE data; NEE processing failure

The correct invited-author count for these sites under the locked rubric is 8 (≥ 21 yr data, ≤ 2 yr latency) — the maximum on the rubric. They were previously allocated based on a snapshot-span estimate that happened to give the same result in most cases, but the presence file now provides an auditable, data-derived basis for the assignment.

The US-WCr investigation (previously on the open issues list) is resolved. US-WCr had a 27-year snapshot span but zero annual NEE_VUT_REF records. Investigation confirmed 24 years of verified presence under the all-variable definition; the NEE absence is a downstream processing limitation, not missing data.

Net effect on headline: +28 authors (2,877 $\rightarrow$ 2,905). The majority of this shift reflects sites whose years_of_data changed when measured against the broader definition rather than a snapshot span.

Headline number

2,905 invited authors across 716 sites from the Apr-28 2026 FLUXNET Shuttle snapshot. All 716 sites are presence-file verified. No snapshot fallback used.

Per-network breakdown

Sorted by site count. All sites are presence-file verified (no fallback).

Network	Sites	Invited authors
AMF	340	1,361
EUF	118	465
ICOS	79	432
JPF	52	179
TERN	52	221
CNF	31	111
KOF	21	53
FLX	18	63
SAEON	5	20
TOTAL	716	2,905

One open caveat before finalising the author list

Caveat 1 — 5-year rubric row: +50 authors if moved to “6–10” tier

The rubric labels “<5 years” and “6–10 years” leave a gap at exactly 5. The current implementation assigns 5-year sites to the <5 row (lower count). 50 sites are affected; the per-network impact under the alternative assignment (5 years $\rightarrow$ “6–10” row) is:

Network	5-yr sites	Swing
AMF	22	+22
ICOS	8	+8
EUF	6	+6
JPF	3	+3
CNF	3	+3
KOF	2	+2
FLX	2	+2
SAEON	2	+2
TERN	2	+2
TOTAL	50	+50

The swing is +1 author per site (uniform across all latency tiers, because the locked rubric’s row-to-row step is uniformly 1). The rubric label needs to be resolved — either change “<5 years” to “≤5 years” (keep current mapping) or “6–10 years” to “5–10 years” (move 5-year sites up). This was deferred at the 2026-05-07 team meeting and **must be resolved before the author list is sent to networks**.

Pending followup task

- **FOLLOWUP-A** (team decision required before sending invitations): Resolve the 5-year rubric row label. +50 authors at stake. See Caveat 1.

Resolved since initial report:

- **~~FOLLOWUP-C:** Regenerate `site_year_data_presence.csv` for all 716 Shuttle sites. **~~ Done 2026-05-12** — all 716 sites now presence-file verified under the all-variable definition.
- **~~FOLLOWUP-E1:** Rebuild the presence indicator to cover all variables, not just NEE. **~~ Done 2026-05-12** — 12-column union (NEE/GPP/RECO VUT+CUT, NT+DT; LE_F_MDS; H_F_MDS).
- **~~FOLLOWUP-E2:** Add month counts to the presence file to enable full-year vs. partial-year classification. **~~ Done 2026-05-12** — `n_months_present` column added; `years_full_year` and `years_partial_year` now computed per site.